Replication of Study 2 by Eitam, Hassin, & Schul (2008, Psychological Science)

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Introduction

Nonconscious goal pursuit (NCGP), although helpful in facilitating performance to achieve a goal, has typically been considered to involve unconscious, nonintentional, inflexible, reproductive, and perceptual-based processing. This stands in sharp contrast to the more intentional, rule-based, conscious, deliberate and flexible processing believed to be at the root of intentional efforts to solve problems and reach goals. These two types of goal pursuits not only involve these different processes, but are also seen as distinct and independent. In addition, because NCGP involves nonconscious reproductions of past associations, it is assumed that the benefits for experiencing NCGP are limited to actions that are consistent with the nonconscious goal (e.g., when the goal of dieting is primed, the resulting action will be to reduce food intake).

Eitam, Hassin, and Schul (2008) studied whether or not NCGP could extend beyond reproductions of past associations and facilitate novel learning, specifically implicit learning. In two experiments, Eitam and colleagues demonstrated that NCGP, accomplished through implicit priming with goal-directed words, facilitated the learning on implicit learning tasks even when the NCGP priming words were not necessary or needed for accurate performance on the implicit learning task. An implicit learning task is one where individuals learn something without intending to learn it and without awareness that they have learned.

The purpose of the present study was to replicate experiment 2 of Eitam, Hassin, and Schul (2008) to examine whether or not the effect of NCGP on implicit learning would be supported; a power of .80 to detect the original effect size was planned. As this is a replication that is part of the Reproducibility Project: Psychology, which asks researchers to replicate the final study in the original article, we replicated experiment 2. In order to replicate the procedures as closely as possible, we asked the authors for their materials; all materials requested were graciously provided. Because we had research assistants, a written protocol was created for the present study for the research assistants to follow; it was unclear from the original article and communication with the original authors whether or not this existed for the original study.

Based on details provided in the paper and through communication with the lead author, all methods were followed as closely as possible to the original. The materials were translated from Hebrew to English, so it is possible that there were slight differences in the materials. For example, because of the translation from Hebrew to English, our priming word-search puzzles were 11 x 11 arrays of letters as opposed to the 10 x 10 used in the original study. The two samples were selected from Universities in different countries: Israel in the original compared to the United States in the replication. The samples were also different in that the sample from the original study was somewhat older, with a mean age of 23.9 years, than the participants in the replication study who had a mean age of 19.65 (*SD* = 1.1), and the proportion of female participants was higher in the replication study (74.8% vs. 60.5%).

Method

The method section has been organized to follow the standard template of the Reproducibility Project: Psychology. Thus, the proposed methods are presented first, followed by a section detailing differences between the proposed methods and those of the original study (Differences from Original Study section), followed by a section detailing differences between the proposed methods and those actually used (Differences from pre-data collection methods plan section).

Power Analysis

The effect size for the finding that has been targeted for replication is a Cohen's *d* of .451, which was the effect size found in the original study (the Reproducibility Project: Psychology guidelines specify using the original effect size when computing power). Consistent with the original study, a two-tailed test with an alpha of .05 will be used. Assuming an equal number of participants in each group, a sample size of 158 participants are needed to achieve a power of 80% to detect an effect this large or larger. For 90% power, 210 participants would be necessary, and for 95% power, 258 participants would be necessary. Given the size of the undergraduate pool that the sample will be drawn from, a sample to achieve 80% power is a more realistic goal especially if the aim is to have data collection completed in a year or less.

Planned Sample

Using a planned power of .80, the power calculation indicated that a total of 158 participants, approximately 79 per group, were required for the replication study. As with the original study, there are certain conditions under which participants' data will be excluded from analysis (detailed below). Thus, participants will be run until 79 in each group have provided usable data. As with the original study, participants were

undergraduate university students recruited from an Introduction to Psychology course pool of volunteers who received course credit for participation, or were from the general student body who received monetary compensation for participation. Students were recruited from Loyola University Maryland, a private, liberal-arts school that is located in the Mid-Atlantic region of the United States. There are approximately 4,000 undergraduates and roughly 60% are female. The ethnic distribution of the students as determined by self-report is approximately: 78% Caucasian, 10% Hispanic, 4% African American, 3% Asian, 3% two or more ethnicities, less than 1% American Indian / Alaskan Native, and less than 1% Pacific Islander / Hawaiian Native. The majority of students, roughly 73%, come from the Middle States region, which includes Delaware, the District of Columbia, Maryland, New Jersey, New York, and Pennsylvania.

Materials

Goal priming task. The UCGP priming task was a word-search puzzle. An 11 x 11 array of letters comprised the word-search task, within which the participants were asked to locate 13 target words. Six words were the same for both conditions (*blue, building, candle, chair, stamp,* and *tree*). For the goal-directed condition seven of the words were associated with achievement (*ambitious, competition, excellence, first, goal, racing,* and *win*). For the control condition seven motivationally neutral words of approximately the same length were used instead (*carpeting*, *diamonds*, *farm*, *hat, limestone, table, and window*). Word-search puzzles for each condition are shown in the Appendix.

SRT task. A widely used incidental-learning task, the serial reaction time task (SRT; Nissen & Bullemer, 1987) was used as a measure if implicit learning. It is widely accepted that participants can demonstrate significant learning of complex sequences in this task

without intention or awareness of the predictability of the sequences. As described in detail below, reaction time is used as an indicator of incidental learning by comparing reaction times during sequences that participants were previously 'trained' on to reaction times during random sequences.

Awareness probes. Several probes were conducted to assess the participants' awareness of the priming task and their explicit knowledge of the SRT task. Explicit knowledge of the SRT stimulus sequence was assessed by asking the participants to describe the sequence in their own words. Then, after being informed that the sequence of stimuli in the SRT task was fixed, the participants were asked to graphically generate a sequence of 10 locations that would follow the same fixed sequence. The responses were used as indicators as the level of explicit knowledge of the sequence ranging from 0 (no knowledge) to 10 (full knowledge) using a computerized scoring algorithm.

Three additional items assessed the participants' achievement motivation. Specifically they were asked how important it was for them to respond quickly and accurately, and a third question asked how important it was that they succeed at the task. A fourth item assessed their overall motivation to learn the sequence in the SRT task. "All ratings were made on a 9-point scale ranging from 1 (*not at all important*) to 9 (*very important*)" (Eitam et al., 2008, p.264).

"Finally, three items assessed whether participants believed the cover story, and whether they had any suspicion regarding the priming manipulation. First, they were asked to speculate, in their own words, on 'the purpose of the word-search task' (which served as the priming manipulation). Next, they were asked 'whether there was anything noteworthy in the word-search task," and last, they were asked directly whether the word-search task

had affected their performance on the SRT task in any way" (Eitam et al., 2008, p.263). Post-experimental questionnaire items are shown in the Appendix.

Procedure

All participants provided informed consent prior to participation; please see the Appendix below for the informed consent form. "Upon arriving at the lab, participants were informed that the experimental session consisted of two 'unrelated experiments.' The first was introduced as a pilot test in which participants were asked to perform a word-search task" (Eitam et al., 2008, p.262). Block randomization was used in order to randomly assign participants to either the goal-priming or control condition. Participants were given an 11 x 11 letter array "in which they had to search for 13 target words. In the goal-priming condition, 7 of these words were associated with achievement" (Eitam et al., 2008, p. 262). "In the control condition, these words were replaced by motivationally neutral ones" (Eitam et al., 2008, p. 262). The words used are listed above. "After completing this task, participants in both groups were thanked and introduced to what was described as the second experiment, in which they were given an implicit-learning task" (Eitam et al., 2008, p. 262).

SRT task. DirectRT software was used to deliver the SRT task. "In each trial of the SRT task, participants were shown a circle that appeared in one of four possible locations on a horizontal line on a computer screen. They were instructed to indicate the circle's location by pressing one of four keys ('3,' '5,' '7,' or '9'). The circle disappeared immediately after the response, and the next circle appeared 200 ms later. An error was signaled by a red 'X' that appeared in the center of the screen and remained until the participant responded correctly" (Eitam et al., 2008, p. 264).

"The task began with a practice block in which the location of the circle was randomly determined. The purpose of this block was to familiarize participants with the task and to discourage them from attempting to learn the sequence explicitly (cf. Reed & Johnson, 1994; Seger, 1997). Next, participants underwent four training blocks. Critically, in these blocks, the sequence of locations in which the circle appeared was fixed. (if '1' indicates the left-most location and '4' the right-most, the sequence was 1324323432; see Stadler, 1992). This sequence was repeated five times per block. All blocks were 50 trials long" (Eitam et al., 2008, p. 264).

"Following the four training blocks, participants engaged in three test blocks (50 trials each). In the first and third blocks (henceforth, random blocks), the locations of the circle were randomly determined" (Eitam et al., 2008, p. 264). There were two constraints for the location of the circle on the random blocks. First, the circle appeared at the same frequency in each location in the random blocks as in the fixed-sequence blocks. Second, the circle never consecutively appeared in the same location. "In the second test block, however, locations followed the fixed sequence used during learning (cf. Seger, 1997)" (Eitam et al., 2008, p. 264).

"In this paradigm, participants are instructed to merely react to the individual circles. Nothing in the instructions or the explicit structure of the task indicates the existence of a learnable underlying structure. Hence, learning in this paradigm is considered unintentional" (Eitam et al., 2008, p. 264).

Awareness probes. "After participants completed the SRT task, we assessed their awareness and explicit knowledge of the learned sequence. First, they were asked to describe the sequence in their own words (Seger, 1997). These verbal descriptions were

scored using slight modifications of Seger's (1997) scoring criteria. Then, after being informed that the locations were displayed in a fixed sequence, participants were requested to generate a sequence of 10 locations that would follow the fixed sequence (cf. Destrebecqz & Cleeremans, 2001). These locations were indicated graphically. A computerized scoring algorithm assessed the level of fit between each of the generated sequences and the actual sequence used in the training blocks, yielding a knowledge score that ranged from 0 (no knowledge) to 10 (full knowledge)" (Eitam et al., 2008, p. 264).

"Three items assessed participants' explicit achievement motivation. First, they were asked to rate how important it was for them to respond quickly. Second, they were asked to rate how important it was for them to respond accurately. A third item asked participants how important it was for them to succeed. A fourth item assessed participants' intention to learn the sequence, asking how important it was for them to 'learn the rule by which the stimulus locations were determined.' All ratings were made on a 9-point scale ranging from 1 (*not at all important*) to 9 (*very important*)" (Eitam et al., 2008, p. 264).

"Finally, three items assessed whether participants believed the cover story, and whether they had any suspicion regarding the priming manipulation. First, they were asked to speculate, in their own words, on the 'purpose of the word-search task' (which served as the priming manipulation). Next, they were asked 'whether there was anything noteworthy in the word-search task,' and last, they were asked directly whether the word-search task had affected their performance" (Eitam et al., 2008, p. 263) on the SRT task.

Analysis Plan

Participants' data were excluded from the analyses if they demonstrated awareness of the motivation link between the goal-priming word-search task and the SRT task. In

addition, participants' whose learning performance on the SRT task was 4 standard deviations from the mean were also excluded.

SRT learning. A difference score was computed between the average of the two random blocks and the one fixed block. The mean difference score for the control condition was then compared to the mean difference score for the experimental condition using an independent samples *t*-test. Likewise, to assure that the differences observed between the control and experimental condition was due to improvements in learning the fixed sequence, the means for the two random blocks of trials were compared for the two conditions using an independent samples *t*-test. Finally, as done in the original study, mean reaction times for the experimental and control conditions were compared regardless of type of test block.

Explicit motivation. A series of *t*-tests were conducted to determine if the control and experimental groups differed on their explicit motivation when performing the SRT task. Specifically, the mean answers on the post-experimental questionnaire about performing the SRT task accurately, quickly, and general desire to succeed at the task were compared with a *t*-test. Intention to learn the fixed sequence was also compared between the two groups with a *t*-test. An overall MANOVA also was conducted using the above questions to detect any motivational differences between the experimental and control groups.

Explicit knowledge of the SRT task. Additional *t*-test analyses were conducted for the two measures of explicit knowledge about the SRT task, specifically the degree of accuracy of describing the SRT task in their own words, and the accuracy of their graphic predictions of the fixed sequence in the SRT task.

Differences from Original Study

Sample. The primary difference between the sample used in the replication study and that used in the original study is that the two samples were selected from different countries: Israel in the original compared to the United States in the replication. Thus, the primary language of participants in the original study was Hebrew, whereas the primary language for participants in the replication was English.

It is not yet known as the replication has not been completed, but it is likely that while the gender distribution (60% female, 40% male) will be consistent with the replication sample, it is likely that the replication sample will be younger on average than the original sample (mean age of 23.9 years). It is anticipated that the mean age for the replication study will be approximately 19 to 19.5 years of age (see *Actual Sample* below for more details about the demographics of the actual replication sample).

Materials & Procedures. The materials used in the original study were in Hebrew rather than English. A translation from Hebrew to English was conducted by a fluent Hebrew-speaking assistant to the replication researchers.

Given that the priming effect in this study is primarily semantic in nature (creating achievement motivation through exposure to words like “win” and “excellence”) a match in meaning between the original Hebrew words and the English translation, taking into consideration cultural differences, was paramount. For example, one of the words in the original study was “race” which in the United States is equally likely to have the meaning “foot race” or “race/ethnicity.” Therefore the translated word “racing” was used instead to prompt the achievement-associated meaning of the word for the participants. In most cases, the exact translation of the word was used, such as in the case of “first,” or a slight

syntactic variation in the word was used, i.e. “win” rather than “to win.” Also, the neutral words were slightly modified in order to have similar length words in the priming and neutral conditions. Specifically, “carpeting” was used instead of “carpet”, “diamonds” was used instead of “diamond”, and “limestone” was used instead of “topaz”.

Because of the use of English translations, one of the priming words, “competition,” had more than 10 letters. Therefore the word-search task had to be enlarged to 11 x 11 letters instead of the 10 x 10 letter task used in the original study.

Additional Analyses

Although not part of the reproducibility project, the authors would like to include additional analyses in an a priori manner. Specifically, the design of this study lends itself well to using a repeated-measures ANOVA to test the effect of the manipulation on implicit learning (as compared to the multiple t-tests used in the original manuscript). Therefore, the authors also propose to examine the impact of the manipulation on implicit learning by using a 2 (condition: experimental vs. control) x 2 (block type: fixed vs. random) ANOVA looking at response times, where condition is a between-subjects factor and block type is a within-subjects factor. The question under investigation would be examined by testing the interaction of condition and block type: one would expect a significant interaction such that those in the experimental condition show a significantly larger decrease in response time when comparing random to fixed blocks than those in the control condition. If the interactive effect is as large as the observed effect size in the original manuscript ($d = .451$ translates to an effect $f = .226$), then for an alpha of 0.05, a correlation among measures of 0 (most conservative possible), and a nonsphericity correction of 1, the planned sample size of 158 would yield a power of 99.5%.

Post Data Collection Methods Addendum

Actual Sample. It was noted in the planned sample section above that participants would continue to be run until 79 in each group provided usable data. Because the data was not scored and analyzed during the data collection time period, and because those scoring the data were blinded to condition, data was collected for 206 participants with the expectation that this would provide at least 79 usable cases in each condition and the first 79 usable cases in each condition would be used for the analyses in order to provide a power of .80 based on the original effect size.

Of the first 79 participants run in the goal-priming condition, 5 had to be excluded. One participant had to be excluded because it was suspected that they were aware of the motivation link between the goal-priming word-search task and the SRT task. The other four participants had to be excluded for technical reasons (see below for details). Thus, goal-priming participants 80 through 84 were included in the sample. Of the first 79 participants run in the control condition, 4 had to be excluded for technical reasons. Thus, control participants 80 through 83 were included in the sample. Two participants, one in each condition, had SRT learning scores (-486.5 and -472.9) that were 4 or more standard deviations from the mean of the SRT learning scores of the 158 participants (M = -33.4, SD = 72.5); as done in the original paper, negative numbers indicate learning: faster times in the fixed sequence block relative to the two random blocks. The next participant in the goal-priming condition had technical difficulties so the subsequent participant was included to bring the sample size of the goal-priming condition to 79. The next participant in the control condition was included to bring the control condition sample size to 79. Thus, data was used for 79 of the first 86 participants in the goal-priming condition; 5 were

excluded for technical reasons, 1 was excluded because it was suspected that they were aware of the motivation link between the goal-priming word-search task and the SRT task, and 1 was excluded because their SRT learning score was 4 standard deviations from the mean of the SRT learning scores. Data was used for 79 of the first 84 participants in the control condition; 4 were excluded for technical reasons and 1 was excluded because their SRT learning score was 4 standard deviations from the mean of the SRT learning scores.

Of the final sample of 158 participants used in the analysis, gender and age data were only collected from 115 (72.8%). This is because this data was not originally assessed as part of the procedure. When the lead researchers realized that this information was not being assessed (after the first 115 participants had already been run), questions to assess gender and age were added to the very end of the procedure (so that subsequent participants' experience of the remainder of the experiment would be no different than the first 115 participants). After IRB approval was obtained, gender and age data was sought from the first 115 participants by sending them a follow-up email. However, not all participants responded to provide this information. For the 115 of 158 who did provide gender and age data, 74.8% (n = 86) were female and the mean age was 19.65 (SD = 1.1).

Differences from pre-data collection methods plan. Although the method plan did not explicitly state that participant age and gender would be assessed, it could be inferred that they would be assessed by the fact that age and gender of the sample was discussed. However, as noted above, this information was not being collected and the lead researchers did not realize it until after the first 115 participants had been run. At this point, questions to assess gender and age were added to the very end of the procedure (so that subsequent participants' experience of the remainder of the experiment would be no

different from the first 115 participants). After IRB approval was obtained, gender and age data was sought from the first 115 participants by sending them a follow-up email.

Although 'technical reasons' were not given as exclusionary criteria in the original methods, it was deemed necessary to include it after reviewing the data. Of all those excluded for technical reasons, the majority of participants excluded did not have response time data for the SRT task because the task shut down before the relevant blocks were recorded. Therefore, they could not be included in the key analysis (SRT learning) because they did not have data for this variable. One participant did have data for these blocks, but was excluded because they had already run part-way through the SRT task before it shut down. After the task shut down, the research assistant restarted the task at the beginning. Therefore, this participant's learning history would be different than all of the other participants (they underwent more training blocks than other participants).

A couple of the variables assessed required subjective assessment of the responses provided by participants (e.g., knowledge of the motivation link between the word-search task and the SRT task, degree of accuracy of describing the SRT sequence in their own words). The original method section did not provide detail as to how these variables would be assessed. The scoring protocol for these (and other) variables is included below in the appendix (i.e., scoring criteria for goal pursuit research project). These variables were originally scored by research assistants who were blind to condition during scoring. The variables assessing suspected knowledge of the motivation link between the word-search task and the SRT task were then verified by the two primary researchers who were also blind to condition while scoring participant responses.

Results

Data preparation

As indicated above, the index of SRT learning was calculated by subtracting the average response time of the two random blocks from the average response time from the one fixed sequence block. Thus, larger negative difference scores are indicative of greater learning. An index of SRT learning using the median, rather than mean, response times was also calculated because the original article indicated that an analysis using the median response times produced an identical pattern of results to that of the mean response times.

Confirmatory analysis

SRT learning. Figure 1 presents the mean RTs of correct responses in the three test blocks. Typically, only correct responses are used to assess learning differences; although it is not explicitly stated in the original manuscript, it was confirmed with the authors that this is what was done in the original study. The average learning score for those in the goal priming condition (M = -22.3, SD = 51.3) was not significantly different from those in the control condition (M = -33.1, SD = 52.1), $t(156) = 1.318$, $p = .189$, d = -0.21. When averaging across all participants and all three blocks, the average accuracy was 93.3%: out of 150 trials, the mean number of correct trials was 139.96 (SD = 8.65). The average accuracy did not differ between groups for any of the three blocks, all ts(156) < 1.71.

Although the average SRT learning score for those in the goal priming condition was not significantly different from those in the control condition, response time performance in the two random blocks was still compared, as done in the original study, to determine if one condition showed enhanced performance in general (not necessarily related to learning). No significant differences were found between the goal priming and control conditions when using either mean or median response times, all ts(156) < 1.25.

Explicit motivation. The average score for explicit motivation to perform the SRT task accurately did not differ significantly between the goal priming condition (M = 7.6, SD = 1.3) and the control condition (M = 7.7, SD = 1.1), $t(156) = -.514$, $p = .608$, $d = 0.08$. The average score for explicit motivation to perform the SRT task quickly did not differ significantly between the goal priming condition (M = 7.2, SD = 1.4) and the control condition (M = 7.2, SD = 1.4), $t(156) = -.229$, $p = .819$, $d = 0.04$. The average score for explicit motivation to generally succeed at the task did not differ significantly between the goal priming condition (M = 7.3, SD = 1.5) and the control condition (M = 7.4, SD = 1.3), $t(156) = -.409$, $p = .683$, $d = 0.07$. The average score for explicit motivation to learn the fixed sequence did not differ significantly between the goal priming condition (M = 5.5, SD = 2.5) and the control condition (M = 5.0, SD = 2.7), $t(156) = 1.354$, $p = .178$, $d = -0.22$.

Explicit knowledge of the SRT task. Explicit knowledge of the fixed sequence order as measured by the degree of accuracy in describing the SRT task in their own words did not differ significantly between the goal priming condition (M = 1.9, SD = 1.2) and the control condition (M = 2.1, SD = 1.2), $t(156) = -0.668$, $p = .505$, $d = 0.11$. Explicit knowledge of the fixed sequence order as measured by their graphic predictions of the fixed sequence in the SRT task almost reached the level of a significant difference when comparing the goal priming condition (M = 4.0, SD = 1.9) and the control condition (M = 3.4, SD = 2.0), $t(156) = 1.949$, $p = .053$, $d = -0.31$.

Exploratory analyses

SRT learning was examined using a 2 (condition: goal priming vs. control) x 2 (block type: fixed vs. random) ANOVA looking at response times, where condition is a between-subjects factor and block type is a within-subjects factor. There was not a significant main

effect of condition, $F(1, 156) = 0.85$, $p = .359$, $\eta^2 = .01$. Not surprisingly, there was a large, significant effect of block type on response times, $F(2, 312) = 20.14$, $p < .001$, $\eta^2 = .11$. Examination of this effect of block type revealed that the average response time during the two random blocks was significantly slower than that of the fixed sequence block, $F(1, 156) = 45.35$, $p < .001$, $\eta^2 = .23$; in other words, implicit learning had occurred for participants in both conditions. However, the hypothesized block by condition interaction was not significant, $F(2, 312) = 0.88$, $p = .42$, $\eta^2 = .01$.

Discussion

Summary of Replication Attempt

The primary result from the confirmatory analysis was that NCGP, accomplished through implicit priming with goal-directed words, did not facilitate the learning of a complex implicit learning task. There was not a significant difference in implicit learning between those who were primed with goal-directed words and those who were not. In fact, the direction of the non-significant effect found was opposite to that of the original study. Thus, the present study failed to replicate the original result.

Commentary

The primary result from the original study was not replicated. The present study *did* replicate the long-standing finding of the learning of complex patterns implicitly without intention or awareness. As with the original study, both groups demonstrated significant implicit learning as indicated by significantly faster mean response times in the fixed blocks compared to the random blocks (although the original paper did not explicitly test this, the data provided was used to perform t-tests revealing that both groups in the original paper also demonstrated significant implicit learning). However, achievement

priming using goal-directed words provided no significant benefit to the implicit learning. Therefore the primary finding from the original study was not replicated. In fact, although not statistically significant, the mean learning score for the control group was larger than that for the priming group. There are, however, several differences in the original study and this study that may account for the failure to replicate the primary result of the original study.

The most glaring difference between the original study and the replication study – that may have been a factor in the failure to replicate the primary finding of the original study – was culture. The original study was conducted in Israel with participants from that country and with stimulus materials presented in Hebrew – probably the participants' native language. Although the replication authors used a translator fluent in both Hebrew and English languages, these linguistic and cultural differences are still potentially significant factors in moderating the results of this replication study. Even though every effort was made to assure equivalency between the original Hebrew words and the English words used in the replication, it is still possible that the cultural meaning of the achievement words was more profound in the context of the Hebrew language in the Israeli culture relative to the English language in the American culture.

Similarly, although participants in both studies were enrolled in an institution of higher learning, the sample from the original study was somewhat older (M = 23.9 years) than the participants in the replication study (M = 19.65, *SD* = 1.1). In addition, the proportion of female participants was higher in the replication study (74.8% vs. 60.5%). Although implicit learning and priming are typically less affected by demographic variables, such as age and gender, it is possible that the cultural differences described

above and these other demographic variables together could be important moderating factors that affected the failure to replicate the primary finding from the original study. For example, given that all participants were students in an institution of higher learning, being older, and possibly more mature and/or focused, the participants in the original study may have been more affected by the priming condition than those in the replication study.

In summary, the replication of significant implicit learning measured with a reaction time task is consistent with a plethora of studies in the literature examining implicit learning. These results again demonstrate that humans are capable of acquiring and applying complex knowledge without intention or awareness. However, the failure to replicate the goal-directed priming effect from the original study raises some interesting questions. It is possible that such affects are, relative to the general ubiquitous implicit learning effect, more readily affected by cultural or demographic variables, such as language, culture-of-origin, or age. Hence, the participants in the replication study differed enough on these factors to demonstrate no priming effect, when one was found in the original study.

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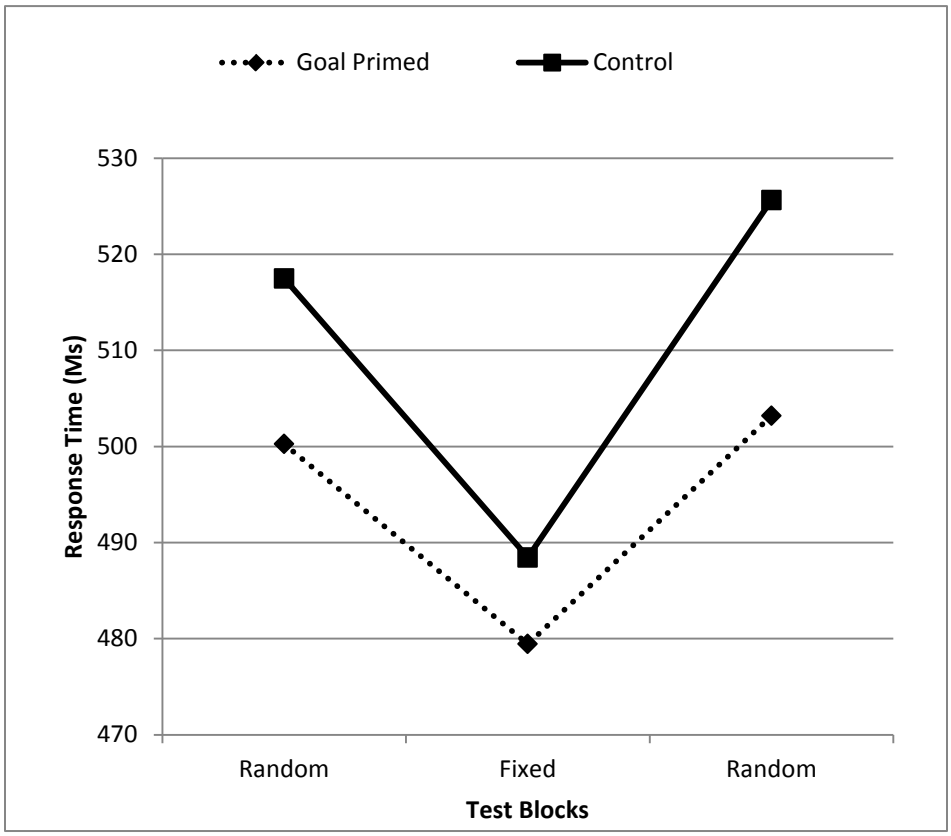
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Figure 1. Response times of the goal-priming and control groups in the three test blocks. The location sequence in the fixed sequence block was identical to that used during training. The location sequence in the random blocks was determined randomly.



Appendix

Word Search Goal-Directed Condition

Word Search Control Condition

Post-experimental questionnaires

Participant Informed Consent Form

Scoring Criteria for Goal Pursuit Research Project

Word Search (Goal-Directed Condition)

Please find the words that appear in the word search below. Words can be placed horizontally, vertically and diagonally. Words can be placed either forwards or back-to-front.

When you find all the words, open the door of the room and the tester will come to give the rest of the instructions.

E E N G G N B G E V O
L X N U N O N D J E V
D C S U O I T I B M A
N E J E C T D L B N N
A L M A F I G L A C I
C L R J I T U J I O W
T E M U R E S Y W U G
S N M U S P M T M N B
V C O F T M C H A I R
E E R T W O Q R K M T
M R Z W G C P Y W O P

Here are the words for the word search:

1. Win
2. Goal
3. Blue
4. Chair
5. Excellence
6. Stamp
7. Ambitious
8. Candle
9. Competition
10. Building
11. First
12. Racing
13. Tree

Word Search (Control Condition)

Please find the words that appear in the word search below. Words can be placed horizontally, vertically and diagonally. Words can be placed either forwards or back-to-front.

When you find all the words, open the door of the room and the tester will come to give the rest of the instructions.

T C E S H S B M N R L
A A L T D L T Y R I N
B R D G U N R A M A O
L P N E T V O E M A F
E E A P A M S M I P V
H T C L H T Q W A P G
A I X H O W O B G I E
V N G N I D L I U B D
R G E F N C H A I R D
R F N I M U Q F W M A
R O W E E R T K P J T

Here are the words for the word search:

1. Hat
2. Farm
3. Blue
4. Chair
5. Limestone
6. Stamp
7. Diamonds
8. Candle
9. Carpeting
10. Building
11. Table
12. Window
13. Tree

Post-experimental Questionnaires

Summary Form for Both Experiments

Thank you for your time and earnestness. Upon completion, it is important for us to know that you answer honestly and seriously the following questions. For the questions below that are followed by numbers, please circle the number that best represents your answer to the question.

How is your mood now?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

very bad very good

The following questions relate to the <u>first</u> experiment (word search):

How hard was the word search for you?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

very easy very hard

To your knowledge what was the goal of this experiment?

_____

Were there specific words that were harder for you to find than others?

_____

Do you have experience with this type of task?

_____

The following questions relate to the <u>second</u> experiment (the circles):

To your knowledge what was the goal of this experiment?

_____

Did the experiment where you searched for words affect your approach to the second experiment? 1. No 2. Yes

Please explain how.

_____

How difficult was the second experiment?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

very easy very hard

How hard did you try during the second experiment?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

I didn't try at all I tried a lot

How important was it to you to respond quickly in the second experiment?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

Not important at all very important

How important was it to you to be accurate in responding to where the circle was?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

Not important at all very important

How important was it to you to succeed in the second experiment?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

Not important at all very important

How quickly did you reply?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

I didn't reply very quickly I replied very quickly

How accurate were you?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

I wasn't accurate at all I was very accurate

In general, how successful were you on the second experiment?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

I wasn't successful at all I was very successful

How important was it to you to figure out the rule that determined where the circles appeared?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

Not important at all very important

How certain were you that there was this 'rule'?

1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9

Sure it did **not** exist sure it did exist

Please call the experimenter to receive the last part of the experiment.

Summary Form Part 2

Now that you know that the circles did appear in their places in a specific pattern / order, what to your knowledge was the length of this pattern? _____

Experiment 2 consisted of 8 different blocks (each block was separated by a break). Please try to guess whether or not the circles appeared in a consistent regular pattern or appeared randomly during the 8 blocks. Please answer by means of writing “Regular” or “Random” below.

Block 1:_____ Block 2:_____ Block 3:_____ Block 4:_____

Block 5:_____ Block 6:_____ Block 7:_____ Block 8:_____

Please describe in words, to the best of your ability, the order that the circles appeared.

_____

_____

_____

_____

The ten rows below represent ten ‘appearances’ of the circles. Each place in the row represents one of the four places upon which the circles appears on the screen. From top to bottom, place one circle in each row according to the pattern in which they appeared, to your knowledge.

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

___ ___ ___ ___

Have you been exposed to any information about goal priming? 1. No 2. Yes

If yes, please explain what you know about goal priming.

_____

_____

Thank you for participating. Please approach the testers to receive a signature or receipt.

Participant Informed Consent Form

Description of Studies. You are being asked to participate in these two research studies because you are a Loyola Maryland University student ***at least 18 years of age***. The first study is a pilot test designed to determine the length of time it takes participants to complete a 'filler' task that will be used in future research. This study involves completing a word search task. The main study is a cognitive study that is investigating college student reaction times. A circle will appear at various places on a computer screen and you will be asked to hit a key that corresponds to the position the circle is in. Afterwards you will be asked some questions about these tasks.

Voluntary Participation. Participation in this study is voluntary. If you decide to withdraw, you may do so at any time without penalty. Even if you withdraw, you will still get research credit for your class or payment of $15 (whichever you are doing the studies for).

Confidentiality. The only identifying information about you (your name) will only be on this informed consent form which will not be connected with your data. Your data (which will not be linked to your name) will be stored on a password protected computer and the informed consent forms (with your name) will be stored in a locked filing cabinet. Only study personnel will know that you have participated in the study.

Risks or Discomforts. It is not expected that participation in either study will be associated with any risks to you beyond what you would experience in you daily life. There is a small possibility that you may become frustrated with the tasks. If they begin to become frustrated and are concerned that continued participation will be upsetting to you, you can discontinue participation at any time.

Benefits. You will receive payment of $15 or course credit / extra credit for your participation. Aside from this, there is no anticipated benefit other than the possibility that you may enjoy completing the tasks and/or will benefit from learning more about the process of participating in research. Information from this study will inform cognitive research.

***Contact Information*:** If you have any questions about this study, or if any problems arise in connection with your participation in the study, you may contact the primary investigators, Dr. Jason Prenoveau at 410-617-5240 (or jmprenoveau@loyola.edu) or Dr. Matthew Kirkhart at 410-617-5498 (or MKirkhart@loyola.edu). If you have any questions about your rights as a research subject you may call the Office of Research and Sponsored Programs of Loyola University Maryland at telephone number 410-617-2561.

THIS PROJECT HAS BEEN REVIEWED AND APPROVED BY THE LOYOLA UNIVERSITY MARYLAND INSTITUTIONAL REVIEW BOARD (PHONE: 410-617-2561).

By signing below, I am indicating that I am a student at Loyola University Maryland, that I am **over 18 years of age**, and that I have read the above information, understand it, and agree to participate in the research study described above.

_____ _____

Subject's Name (printed) and Signature Date

_____ _____

Name (printed) and Signature of Research Assistant or PI Date

SCORING CRITERIA FOR GOAL PURSUIT RESEARCH PROJECT

Check to ensure that folder ID matches ID on all materials in folder.
Check to ensure that folder ID matches excel ID you are entering data for.
Check to ensure that crossword (priming or control) matches condition specified for this participant ID in excel.

Word Search

Enter time in SECONDS.
(Minutes x 60) + seconds = time

Summary Form for Both Experiments

Item/Code	How to Score
Mood	Enter circled value (1-9); Enter 0 if left blank
Hard Search	Enter circled value (1-9); Enter 0 if left blank
Experiment 1 goal	Enter 1 if statement does not meet the criteria for 2 below Enter 2 if statement indicates knowledge that the words in the word search would affect performance in the SRT task
Hard words	Enter 1 if anything but "yes" Enter 2 if "yes"
Experience	Enter 1 if anything but "yes" Enter 2 if "yes"
Experiment 2 goal	Enter 1 if the statement does not meet the criteria for 2 below Enter 2 if the statement indicates that the performance on the SRT task would be affected by the words in the word search
Affect	Enter 1 if circled "No" Enter 2 if circled "Yes"
Explain if yes	Type in the text provided by the participant if they circled "Yes" above
Difficult	Enter circled value (1-9); Enter 0 if left blank
Try	Enter circled value (1-9); Enter 0 if left blank
Quickly	Enter circled value (1-9); Enter 0 if left blank
Accurate	Enter circled value (1-9); Enter 0 if left blank
Succeed	Enter circled value (1-9); Enter 0 if left blank
How quick	Enter circled value (1-9); Enter 0 if left blank
How accurate	Enter circled value (1-9); Enter 0 if left blank
How successful	Enter circled value (1-9); Enter 0 if left blank
How important	Enter circled value (1-9); Enter 0 if left blank
How certain	Enter circled value (1-9); Enter 0 if left blank

Summary Form Part 2

Item/Code	How to Score
Pattern Length	Leave blank
B1	Enter 1 if "Random" or 2 if "Regular"
B2	Enter 1 if "Random" or 2 if "Regular"
B3	Enter 1 if "Random" or 2 if "Regular"
B4	Enter 1 if "Random" or 2 if "Regular"
B5	Enter 1 if "Random" or 2 if "Regular"
B6	Enter 1 if "Random" or 2 if "Regular"
B7	Enter 1 if "Random" or 2 if "Regular"
B8	Enter 1 if "Random" or 2 if "Regular"
Order describe	**Correct sequence = L, RC, LC, R, RC, LC, RC, R, RC, LC or 1, 3, 2, 4, 3, 2, 3, 4, 3, 2** Enter 0 if they mention a false or irrelevant feature or leave blank Enter 1 if they mention "sequence" or "jumps" without any additional elaboration. Enter 2 if they mention two of the base frequencies without any description of actual order (i.e. "most of the dots in the sequence appeared in the left and right middle spaces") Enter 3 if they identified a sequence of three or four correct positions in a row (i.e. "it went space 2, then space 3, then space 4, and then back to space 3 again") Enter 4 if they identified a sequence of five to nine correct positions in a row (i.e. "at one point it went 4, 3, 2, 3, 4, 3, 2 if you numbered the spaces 1 to 4 starting with a 1 position on the far left space") Enter 5 if they identified the correct 10 position sequence
R1	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R2	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space

	Enter -1 if no dot is placed in the space
R3	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R4	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R5	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R6	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R7	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R8	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R9	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
R10	Enter 1 if dot in left-most space, 2 if dot in left-center space, 3 if dot in right-center space, and 4 if dot in right-most space Enter -1 if no dot is placed in the space
Exposed	Enter 1 if circled “No” Enter 2 if circled “Yes”
Text	If “Yes” above, enter text
Birth Date	Enter birth month and year
Date Run	Enter day, month, and year subject was run
Sex	Enter participant sex (F = 1; M = 2)